Smart Library Request Workflow in ServiceNow

A comprehensive guide to implementing a role-based Library Management System

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1. Problem Statement & Objective

Problem Statement

In many academic institutions, managing a library system manually creates challenges such as lack of real-time updates on book availability, inefficient approval

processes, and limited visibility into borrowing patterns. Students often face issues in knowing which books are available, while librarians struggle with tracking requests, approving or rejecting them, and maintaining accurate records of issued or returned books.

To address these challenges, a role-based Library Management System needs to be implemented on the ServiceNow platform. This system should allow students to request books, while librarians manage the book inventory and approve borrow requests. Automated workflows, role-based access controls, and reports will ensure a streamlined process for both parties.

Objective

The objective of this project is to implement a role-based Library Management System in ServiceNow that simplifies the process of borrowing and managing books. The system allows students to submit borrow requests and track their status, while librarians manage the book catalog and approve or reject requests.

Through automated workflows, book availability is updated in real time, role-based access ensures proper security, and reporting features like "Most Borrowed Books" provide insights for better resource management.

Business Objectives

The main business objective is to create a digital library system that:

- Reduces manual effort
- Improves accuracy in book tracking
- Enhances user experience for both students and librarians
- Ensures efficient operations and transparency
- Provides faster access to resources

2. Phase 1: Requirement Analysis & Planning

The system requires two roles: student and librarian. Two tables are needed: Book (details, status) and Borrow Request (transactions). Workflows will automate approvals and status updates, ACLs will control access, and reports will give insights.

Functional Scope

- **Book Management:** Librarians can add, update, and maintain book details with status tracking (Available, Issued, Lost, Returned).
- **Borrow Requests:** Students can create and track requests; librarians can approve, reject, or close them.
- **Workflow Automation:** Approval of requests will automatically update the book status.
- **Role-Based Access:** Students have limited access to requests, while librarians manage the entire system.
- **Reporting:** Generate insights such as "Most Borrowed Books" for library usage analysis.

Execution Roadmap

1. **Role Setup:** Create student and librarian roles.
2. **Table Design:** Build Book and Borrow Request tables with required fields.
3. **Form Configuration:** Add fields, set default values, and establish related lists.
4. **Workflow Automation:** Configure Flow Designer for approval and book status updates.
5. **Access Control:** Define ACLs for role-based permissions.
6. **UI Enhancements:** Apply UI Policies for form restrictions and usability.
7. **Reporting:** Create dashboards/reports like Most Borrowed Books.

8. **Testing & Validation:** Use sample data, impersonation, and debug tools to verify end-to-end flow.

Milestones

- Milestone 1 Define roles: student and librarian.
- Milestone 2 Create Book (u_book) and Borrow Request (u_borrow_request) tables.
- Milestone 3 Add fields to Book: Title, Author, ISBN, Status (Available, Issued, Lost).
- Milestone 4 Add fields to Borrow Request: Requested By, Book, Request Date, Status.
- Milestone 5 Show Borrow Requests on Book (Related List).
- Milestone 6 Prevent Requesting Already-Issued Books (Reference Qualifier).
- Milestone 7 Flow Designer – Approve → Issue Book.
- Milestone 8 UI Policy: Lock fields when approved.
- Milestone 9 Access Control (ACLs) for roles.
- Milestone 10 Report – "Most Borrowed Books".

3. Phase 2: Backend Development & Configurations

The backend setup focuses on creating roles, tables, and automation to manage the library process efficiently. A reference qualifier ensures only available books can be requested, and Flow Designer automates approvals.

Data Architecture Setup

Creation of Roles

1. Open ServiceNow > User Administration > Roles.
2. Create Role: **librarian**.
3. Create Role: **student**.

Creation of Tables

1. Navigate to System Definition > Tables.
2. Create Table: **Book** (Label), `u_book` (Name).
3. Create Table: **Borrow Request** (Label), `u_borrow_request` (Name).

Adding Fields to Book Table

- 1. **Title:** String
- 2. **Author:** String
- 3. **ISBN:** String
- 4. **Status:** Choice (Available, Issued, Lost). Default: Available.

Adding Fields to Borrow Request Table

- 1. **Requested By:** Reference (sys_user)
- 2. **Book:** Reference (u_book)
- 3. **Request Date:** Date/Time

- 4 **Status:** Choice (Requested, Approved, Rejected, Returned). Default: Requested.

Reference Qualifier Configuration

Goal: Prevent requesting already-issued books.

```
Table: Borrow Request [u_borrow_request]
Field: Book
Reference Qualifier Condition: Status is Available
```

4. Phase 3: Development

Flow Designer Automation (Approve → Issue Book)

This flow automates the status updates when a librarian approves a request.

Flow Logic

1. **Trigger:** Borrow Request Created or Updated (Condition: Status = Requested).
2. **Action (Approval):** Ask for Approval from Role: *librarian*.
3. **Logic:**
    - 4 If Approved → Continue flow.
    - 5 If Rejected → Set Status = Rejected.
6. **Action (Update Book):** Set Book Status = *Issued*.
7. **Action (Update Request):** Set Borrow Request Status = *Approved*.

8. **Action (Notification):** Send Email to Requester ("Your Borrow Request has been Approved").

UI Policy Configuration

Policy 1: Make Return Date Mandatory

Short Description: Make Return Date mandatory when Issued.

Condition: Status = Issued

Action: Return Date (Mandatory = True, Visible = True)

Policy 2: Lock Fields on Approval

Condition: Borrow Request Status = Approved

Action: Lock fields (Book, Requested By) to prevent accidental edits.

5. Phase 4: Data Migration, Testing & Security

Access Control (ACLs)

Book Table [u_book]

- **Read:** Student, Librarian

- **Create/Write/Delete:** Librarian only

Borrow Request Table [u_borrow_request]

- **Create:** Student
- **Read:** Student (own records), Librarian (all)
- **Approve/Write/Delete:** Librarian

Reporting

Report: "Most Borrowed Books"

- **Source Type:** Table (Borrow Request)
- **Type:** Bar Chart / Pie Chart
- **Group by:** Book
- **Aggregate:** Count
- **Filter:** Status = Approved

6. Phase 5: Deployment, Documentation & Final Presentation

Troubleshooting & Validation

- **Flow Execution:** Verified end-to-end execution. Checked trigger conditions, approvals, and book issuance updates.

- **Reference Qualifier:** Debugged cases where already-issued books were visible. Corrected conditions to filter only available books.
- **UI Policy:** Verified Return Date mandatory setting and field locking.
- **Access Control:** Confirmed students cannot edit book details or approve requests.

Innovation

- **End-to-End Automation:** Eliminated manual tracking of borrow requests.
- **Optimized Request Handling:** Reference qualifiers reduce errors and duplicate borrow entries.
- **User Experience:** Dynamic forms keep the interface clean for students and librarians.

Future Scalability

- Auto-notifications for overdue books via SMS/Email.
- Integration with external library systems or digital catalog APIs.
- Enhanced reporting dashboards for student activity and overdue analysis.

7. Conclusion

> The Library Management Project successfully streamlined the process of borrowing and managing books within ServiceNow by integrating role-based access, automated approvals, and dynamic workflows.
>
> Students can easily submit borrow requests, while librarians efficiently review, approve, and issue books with proper validations. The solution ensures transparency through status tracking, notifications, and reporting features. By combining automation with usability, the project delivers a scalable foundation that can be extended to broader academic services in the future.

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